



RAMCO INSTITUTE OF TECHNOLOGY

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NBA Accredited UG Programs: CSE, EEE, ECE and MECH

Department of Computer Science and Business systems

Academic Year: 2024- 2025 (Even Semester)

Degree, Semester & Branch: B.Tech., IV Semester & CSBS

Course Code & Title: AL3451 & Machine Learning

Name of the Faculty member (s): Ms.M. Preethi Ram, AP/CSBS

Innovative Practice Description

Unit / Topic: Unit 5 / Bootstrapping

Course Outcome: CO5

Activity Chosen: Learning by Teaching

Justification:

Bootstrapping in Machine Learning refers to the method of resampling data with replacement to estimate model accuracy and reduce variance. The Learning by Teaching strategy engages students in actively learning the concept by preparing to teach it to their peers. In this activity, students are first assigned the topic of bootstrapping, encouraged to research its principles, applications, and benefits. Then, each student or group is tasked with explaining the concept to classmates using examples, visual aids, or mini-presentations. This method promotes a deeper understanding as teaching requires clear articulation and comprehension. It also boosts engagement, builds confidence, and fosters collaborative learning as students clarify doubts, correct errors, and reinforce concepts through peer instruction.

Time Allotted for the Activity: 35 minutes

Details of the Implementation:

- Prior to the session, the instructor gauged students' interest in teaching the concept of Bootstrapping. Several students expressed enthusiasm for this opportunity.
- Based on their interest, students volunteered to present their understanding of Bootstrapping to their peers.
- Adequate preparation time was allotted, with the topic assigned a day before the class to allow students to organize their content and teaching approach.
- During the session, the student presenters delivered their lessons on Bootstrapping, covering key aspects such as the purpose of Bootstrapping and its real-time usage.
- To maintain engagement, the presenters incorporated real-world examples and interactive elements. The class was encouraged to participate actively by posing questions and sharing insights during the presentations. This peer-to-peer interaction fostered a collaborative learning environment, allowing students to explore diverse perspectives and deepen their understanding of Bootstrapping.

CO – PO / PSO mapping:

CO	PO1	PO2	PO3	PO4	PSO1	PSO2
CO 5	2	2	2	1	2	2

(1 – Low 2 – Moderate 3 High)**PO / PSO mapped:**

Innovative practice	PO1	PO2	PO3	PO4	PSO1	PSO2
	2	2	2	1	2	2
Justification for correlation	Students apply mathematical and statistical principles to understand and implement bootstrapping techniques, enhancing their ability to solve complex engineering problems.	By analyzing the effectiveness of bootstrapping in various scenarios, students develop critical thinking skills to assess and formulate solutions to complex engineering problems.	Implementing bootstrapping methods allows students to design robust machine learning models that generalize well to new data, considering public health, safety, and environmental impacts.	Students use bootstrapping to conduct experiments, analyze data, and draw valid conclusions, enhancing their research capabilities in machine learning.	Students apply bootstrapping in conjunction with advanced technologies like AI and ML to solve contemporary problems, demonstrating proficiency in modern tools.	Implementing bootstrapping enhances students' skills in computational statistics and software development, providing solutions to real-world business problems.

Images / Screenshot of the practice



Fig.1. Students listened to the topic delivered by their peers



Fig.2. Jaydeep shared his ideas to his class mates

- **Reflective Critique:**

- ❖ ***Feedback of practice from students and other stakeholders:***

- All the students were eagerly listened the class. Students felt they are able to understand the topic.
- The students were able to remember key details and concepts more easily.
- The students express that role of a teacher boost their confidence in knowledge.
- Students highlighted that this practice helped them improve their communication skills, such as clarity, organization, and the ability to present complex ideas in a simplified manner.

- ❖ ***Benefit of the practice:*** (E.g.: Outcome attainment would have increased due to innovative practice over conventional practice)

- Students actively engaging in teaching their peers gain a more profound understanding about bootstrapping.
- Actively participating in the teaching process helps students remember key details and concepts better compared to passive learning methods
- This activity helped the student to improve communication skills.
- Students spent their time in self-learning.
- This activity encouraged the students to share their knowledge with others.

- ❖ ***Challenges faced in implementation:***

- Effectively motivate students who are not participating in the activity due to shyness, fear of public speaking, or lack of confidence. Motivate the students to overcome their shyness and engage in active learning.
- Implementing Learning by Teaching required additional time compared to traditional teaching methods. Planning and executing interactive teaching sessions was challenging.

References:

1. <https://effectiviology.com/protege-effect-learn-by-teaching>
2. https://www.researchgate.net/publication/351905566_Impact_of_Seminars_on_Student_Soft_Skills_Development